

Shashank J

Senior Software Engineer | Distributed Systems & Platform Architect

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience architecting high-performance distributed systems that bridge the gap between financial compliance and massive scale. I specialize in building resilient, event-driven platforms using Go, Java, and Temporal.io, having recently delivered a core workflow engine processing 20M+ daily transactions. I am focused on productionizing GenAI via Model Context Protocol (MCP) to solve real-world operational bottlenecks and drive significant infrastructure cost savings.

TECHNICAL EXPERTISE

- **Languages:** Java, Go, Python, SQL, Elixir.
- **Distributed Systems:** Temporal.io (Advanced Orchestration), Microservices, Non-Blocking I/O, Messaging (Brokered & Brokerless), System Design, PUB/SUB, Circuit Breaking.
- **GenAI & ML:** RAG (Retrieval Augmentation), MCP (Model Context Protocol), LLM Fine-Tuning, Object Detection, Production-grade AI Integrations, Spring AI.
- **Cloud & Infrastructure:** AWS (ECS, Fargate, Lambda, DynamoDB, Aurora RDS, OpenSearch, CodePipeline), Docker, PostgreSQL, Event-Driven Architectures.

PROFESSIONAL EXPERIENCE

Credit Saison India

Senior Software Engineer

Nov 2021 – Present

Bangalore, India

- **Scalable Orchestration:** Architected an enterprise-grade workflow engine using **Temporal.io** to handle **20M+ daily workflows**. Implemented fault-tolerant patterns and worker auto-scaling that improved execution speed by 90% and enabled end-to-end visibility for complex financial states.
- **Agentic AI Innovation (Aegis):** Spearheaded a configurable Java-based Agentic AI platform for L1/L2 support, continuous funnel monitoring and automated Root Cause Analysis (RCA). By aggregating 30+ data signals over data lakes, configuring playbooks, the platform significantly reduced MTTD/MTTR and **eliminated 15 manual support roles** who were L1/L2.
- **Communication Overhaul:** Re-architected a legacy notification system into a non-blocking, event-driven microservice optimized for high-volume API/DB interactions. Managed a **zero-downtime migration** for 25+ partners while increasing delivery reliability from 94% to 99%.
- **Platform Modernization:** Engineered a provider-agnostic, RBI-compliant Video KYC (VKYC) platform. Developed real-time session tracking and adaptive routing that increased funnel conversion rates from **30% to 60%**.
- **Computer Vision (Oden):** Developed a custom Aadhaar masking solution using Computer Vision with **97% accuracy**, outperforming external vendors (87%). Integrated the solution into real-time loan processing pipelines to secure the entire organization document lake.
- **AI Impact:** Fine-tuned LLMs to automate partner API documentation parsing, reducing integration lead time by 50% across 8 major fintech integrations in 2024.

TCS – Credit Suisse

Software Engineer

Aug 2020 – Oct 2021

Bangalore, India

- Engineered a load-aware database router that dynamically balanced traffic, reducing primary database contention by 35% during peak trading windows.
- Automated the migration of 12M+ privileged accounts with zero data loss using custom Go-based ETL tooling and validation scripts.

ENGINEERING IMPACT & OPEN SOURCE

- **Joyboy (Go/Docker):** Developed an open-source, worker-based container orchestrator providing ECS-like management. Implemented a custom scheduler, task heartbeat monitoring, and a RESTful API for single-node cluster management.

- **Laughingale (Go/PostgreSQL):** Engineered a high-performance caching system to offload read-heavy workloads. Implemented incremental polling strategies to synchronize PostgreSQL data into MongoDB for real-time analytics.

AWARDS & HONORS

- **Einstein Mind Award:** For excellence in Computer Vision & Aadhaar Masking Platform.
- **Outside the Box Thinker:** For architecting the VKYC Automation Platform, saving 200K+ monthly manual audits.

EDUCATION

Sai Vidya Institute of Technology (VTU)
B.E. Computer Science (CGPA: 8.2/10)

2016 – 2020
Bangalore, India